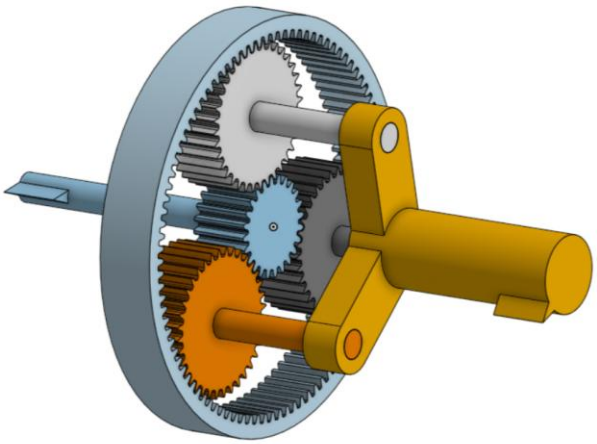
CAD PROJECTS:

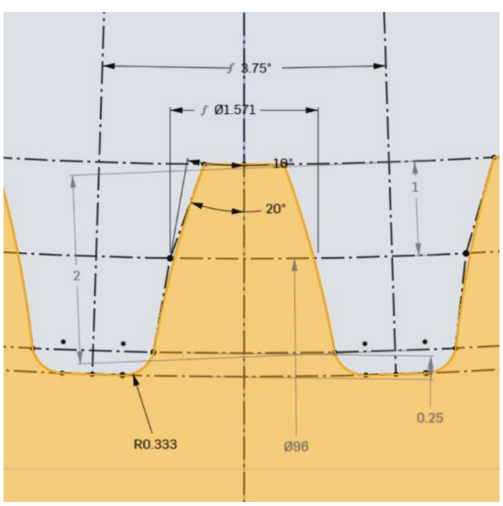
(excluding IP protected TJO and Treace work)

Planetary Gear:



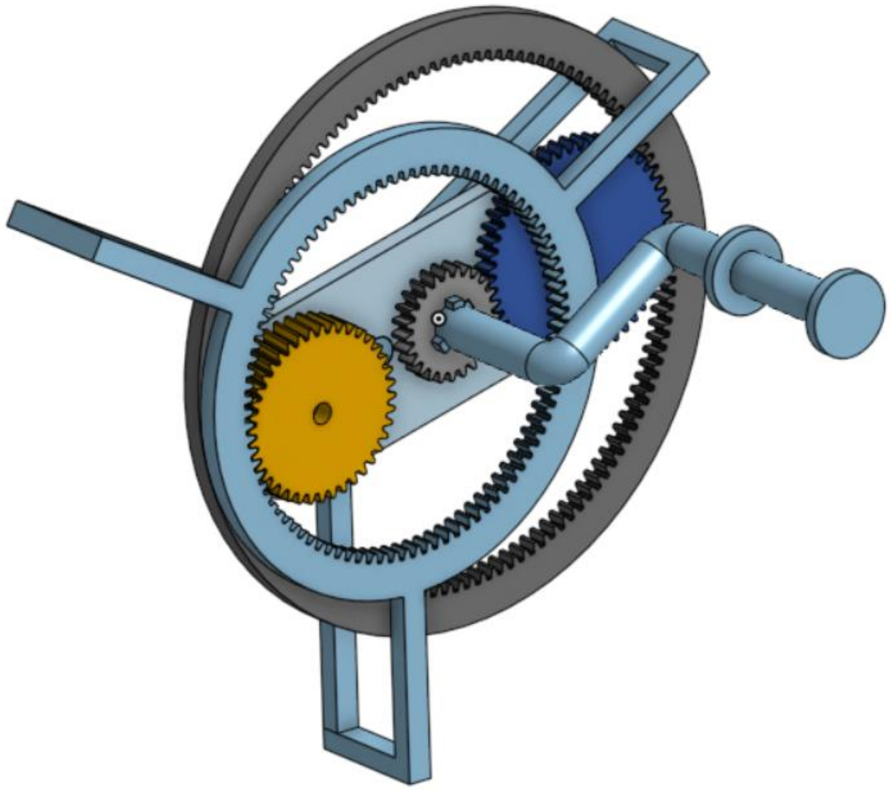
SUN: 24 teeth= N_S
PLANET: 36 teeth= N_P
RING: 96 teeth= N_R
Teeth of ring and sun need to be integers divisible by 3. We desire a gear ratio between the carrier and the sun gear output of 1 to 5.

Gear Ratios:
Orbit Ratio between the Planet gear rotating around the Ring can be modelled by $\frac{N_P}{N_R} = \frac{36}{96} = 0.375$ or for every 1 rotation of a planet gear, the planet makes 0.375 of an orbit around the ring.
The turn ratio between the sun gear and the planet carrier is $\frac{N_S}{N_S+N_R} = \frac{1}{5} = \frac{1}{1+4} = \frac{24}{24+96}$
Ring Gear modelled by hand, spur gears made from an onshape add-on



Teeth for all gears have a module of 1mm, a pitch of pi, a pressure angle $\alpha = 20°$and a root fillet of 1/3. Angle between repeating parts for the ring gear should be $\frac{1}{96} * 360 = 3.75°$
Tooth diameter is reference diameter – 2 mm, and groove diameter is reference diameter +2.5mm for ring gears of pitch module 1mm.

Gear Contraption:



Practice making ring gears, gear chains, and weird mechanisms.

Teeth:

25:50, 50:125, 125:40, 40:90

Sensor Rig for a compound eye camera

Hex lattice patterned across a hemisphere using SolidWorks revolved boss-extrude with circular pattern features. Inspired by compound eyes and icosphere geometry in Blender.

Senior Design Project, a mechanical nerve tensiometer

Assembly includes 7custom parts and 2 off-the-shelf components, with sliding mechanism designed to translate axial tension into a measurable readout